

SEO Machine

Complete Guide

How to Turn Claude AI Into a Full SEO Content Team
For WordPress Agencies Publishing Content at Scale

Based on analysis of github.com/TheCraigHewitt/seomachine

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1. What Is SEO Machine?

SEO Machine is a free, open-source toolkit that turns Claude AI into a full SEO content team. It handles everything from keyword research to writing, optimizing, and publishing articles directly to WordPress - all through simple commands.

Think of it as hiring an SEO strategist, content writer, editor, and publisher - but it runs inside your terminal.

Quick Facts

- **Created by:** Craig Hewitt (CEO of Castos, podcast hosting platform with 40,000+ brands)
- **Cost:** Free and open-source. No hidden fees, no paid tier.
- **GitHub Stars:** ~2,900 stars, 497 forks
- **Requirements:** Claude Code subscription + API keys for analytics (optional)

2. What Problems Does It Solve?

Problem	How SEO Machine Fixes It
Keyword research takes hours	Automated research with real SERP data in minutes
Writers miss SEO best practices	Built-in SEO optimizer agent checks every article
No internal linking strategy	Internal linker agent adds links automatically
Meta tags are an afterthought	Meta creator generates optimized titles & descriptions
Publishing to WordPress is manual	One command publishes with full Yoast SEO metadata
Content quality is inconsistent	11 specialized agents enforce quality standards
Hard to scale beyond 5-10 articles/day	Repeatable pipeline designed for high volume

3. The Complete Architecture

3.1 Three Building Blocks

- **Commands** - What you type to start a workflow (22 available)
- **Agents** - Specialized AI roles that run automatically (11 available)
- **Python Modules** - Analytics and scoring engines (34 available)

3.2 How They Connect

```
YOU type a command (e.g., /write "best podcast hosting")
|
v
COMMAND orchestrates the workflow
|
v
AGENTS run automatically (SEO, meta tags, linking, keywords)
|
v
PYTHON MODULES score and analyze the content
|
v
FINAL ARTICLE ready in drafts/ folder
|
v
/publish-draft sends it to WordPress with full SEO metadata
```

4. Step-by-Step: How Content Gets Made

Step 1: Research (/research [topic])

What happens:

- Analyzes search intent (informational, transactional, commercial)
- Pulls keyword data (volume, difficulty, competition)
- Studies top 10 SERP results for the topic
- Identifies content gaps competitors are missing
- Generates a detailed content brief saved in research/ folder

Step 2: Write (/write [topic])

What happens:

- Uses research brief to create a 2,000-3,000+ word article
- Follows your brand voice and style guide from context/ files
- Structures with proper H2/H3 headings, intro, conclusion
- Formats in WordPress block format (Gutenberg-ready)

Then these 4 agents auto-run on the draft:

- **SEO Optimizer:** Checks title tags, headings, keyword placement, image alt text, URL slug
- **Meta Creator:** Generates 5 meta title/description variations with SERP preview
- **Internal Linker:** Adds relevant links to other pages on your site
- **Keyword Mapper:** Analyzes keyword density and ensures natural placement

Step 3: Optimize (/optimize [file])

Final SEO polish pass:

- Checks readability score (Flesch Reading Ease)
- Compares word count against top 10 competitors
- Validates keyword density (target: 1-2%)
- Ensures proper heading hierarchy
- Checks for missing elements (FAQ, table of contents, etc.)

Step 4: Publish (/publish-draft [file])

Uploads to your WordPress site with:

- Yoast SEO meta title and description
- Focus keyword for Yoast

- Proper categories and tags
- WordPress block format (Gutenberg-ready)

5. All 22 Commands Explained

Core Workflow Commands

Command	What It Does
/research [topic]	Full keyword and competitor research, generates brief
/write [topic]	Creates complete SEO article with auto-optimization
/rewrite [topic]	Updates and improves existing content
/optimize [file]	Final SEO polish before publishing
/publish-draft [file]	Publishes to WordPress with Yoast metadata
/article [topic]	Simplified one-step article creation
/scrub	Removes AI-detectable patterns from content

Research Commands

Command	What It Does
/research-serp [keyword]	Deep SERP analysis for a target keyword
/research-gaps	Finds content gaps vs your competitors
/research-trending	Identifies trending topics in your niche
/research-performance	Analyzes which existing content needs improvement
/research-topics	Discovers topic cluster opportunities

Analytics & Strategy Commands

Command	What It Does
/performance-review	Reviews content performance using GA4 and GSC data
/priorities	Creates a prioritized content roadmap
/cluster [topic]	Builds topic cluster with pillar and supporting articles
/content-calendar	Plans monthly publishing schedule
/analyze-existing [URL]	Audits existing content for improvements

Landing Page Commands

Command	What It Does
---------	--------------

/landing-write	Creates conversion-optimized landing pages
/landing-audit	CRO audit of existing landing pages
/landing-research	Competitor landing page research
/landing-competitor	Deep competitor landing page analysis
/landing-publish	Publishes landing pages to WordPress

6. The 11 AI Agents

Each agent is a specialist that runs automatically when needed:

Agent	Role	Triggered By
SEO Optimizer	Checks on-page SEO (titles, headings, keywords, alt text)	/write
Meta Creator	Generates 5 meta title/description options	/write
Internal Linker	Adds strategic internal links to your pages	/write
Keyword Mapper	Maps keyword placement, checks density	/write
Content Analyzer	Full analysis (intent, readability, length, SEO score)	/analyze-existing
Editor	Makes content sound human and natural	/write
Headline Generator	Creates 10+ headline variations	/write
CRO Analyst	Analyzes conversion optimization	/landing-audit
Landing Optimizer	Full landing page optimization	/landing-write
Performance Agent	Data-driven content prioritization	/performance-review
Cluster Strategist	Organizes topics into clusters	/cluster

7. Data Integrations

Google Analytics 4 (GA4)

- Traffic data (sessions, users, pageviews)
- Engagement metrics (time on page, bounce rate)
- Conversion tracking

Google Search Console (GSC)

- Keyword rankings and positions
- Impressions and click-through rates
- Search query data

DataForSEO

- SERP position tracking
- Keyword volume and difficulty scores
- Competitor keyword data

WordPress REST API

- Direct article publishing from terminal
- Yoast SEO metadata (meta title, description, focus keyword)
- Categories, tags, and featured images

8. The Scoring Systems

SEO Quality Score (0-100)

Comprehensive scoring across: title optimization, heading structure, keyword usage, content length, readability, internal links, and metadata.

Opportunity Score (8 Weighted Factors)

Used to prioritize which content to create or update:

Factor	Weight	What It Measures
Search Volume	25%	Monthly search volume for the keyword
Current Position	20%	Where you currently rank (if at all)
Search Intent	20%	How well your content matches user intent
Competition	15%	How hard it is to rank for this keyword
Cluster Fit	10%	How well it fits your topic clusters
CTR Potential	5%	Expected click-through rate
Freshness	5%	How recently the content was updated
Trend Direction	5%	Is search volume growing or declining

Readability Score

- Flesch Reading Ease score
- Grade level analysis
- Sentence length and complexity checks
- Passive voice detection

9. How This Helps an Agency Publishing 100 Blogs/Day

The Challenge

Publishing 100 WordPress blogs daily requires:

- 100 keyword research briefs
- 100 articles written (200,000 - 300,000 words total)
- 100 SEO optimization passes
- 100 sets of meta tags
- 100 internal linking reviews
- 100 WordPress uploads with Yoast metadata

With humans alone, this needs 20-30 writers, several SEO specialists, and editors.

How SEO Machine Changes This

Phase 1: Batch Research

Run `/research-topics` and `/cluster` to identify 100+ topics at once. Use `/priorities` to rank them by opportunity score. This replaces days of manual keyword research.

Phase 2: Batch Content Creation

For each topic, run `/write [topic]`. Each article gets written (2,000-3,000 words), SEO-optimized automatically by 4 agents, gets meta tags generated, internal links added, and keywords mapped. One operator can produce articles continuously.

Phase 3: Quality Control

Run `/optimize` on each draft for final polish. Use `/analyze-existing` to audit quality. The Python scoring modules give objective quality metrics - no guesswork.

Phase 4: Batch Publishing

Run `/publish-draft` for each article. It uploads to WordPress with full Yoast SEO metadata, proper formatting, categories, and tags.

Realistic Daily Workflow

Time Block	Activity	Output
Morning	Batch research and prioritization	100 topic briefs
Throughout Day	Run <code>/write</code> commands (multiple operators)	100 draft articles
Ongoing	Auto-agents optimize each draft	100 optimized articles

Afternoon	Run /optimize for final pass	100 publish-ready articles
Evening	Run /publish-draft in batch	100 live WordPress posts

What You Still Need

- **Human review:** Spot-check articles for accuracy, especially YMYL topics
- **Brand customization:** Fill out context files once (brand voice, style guide)
- **API keys:** GA4, GSC, DataForSEO for data-driven decisions (optional)
- **WordPress setup:** Install the Yoast REST API plugin (one-time, 5 minutes)
- **Claude Code subscription:** Required to run the commands

Cost Comparison

Approach	Monthly Cost (100 articles/day)
Human writers + SEO team	\$50,000 - \$150,000
AI writing tools (Jasper, Surfer, etc.)	\$5,000 - \$15,000
SEO Machine + Claude Code	Claude subscription + API costs (~\$200-500)

10. Setup in 5 Steps

Step 1: Clone the Repository

```
git clone https://github.com/TheCraigHewitt/seomachine.git
cd seomachine
```

Step 2: Install Python Dependencies

```
pip install -r data_sources/requirements.txt
```

Step 3: Configure API Keys

Copy .env.example to data_sources/config/.env and fill in:

- GA4 Property ID and credentials path
- Google Search Console site URL
- DataForSEO login and password
- WordPress URL, username, and app password

Step 4: Customize Your Context

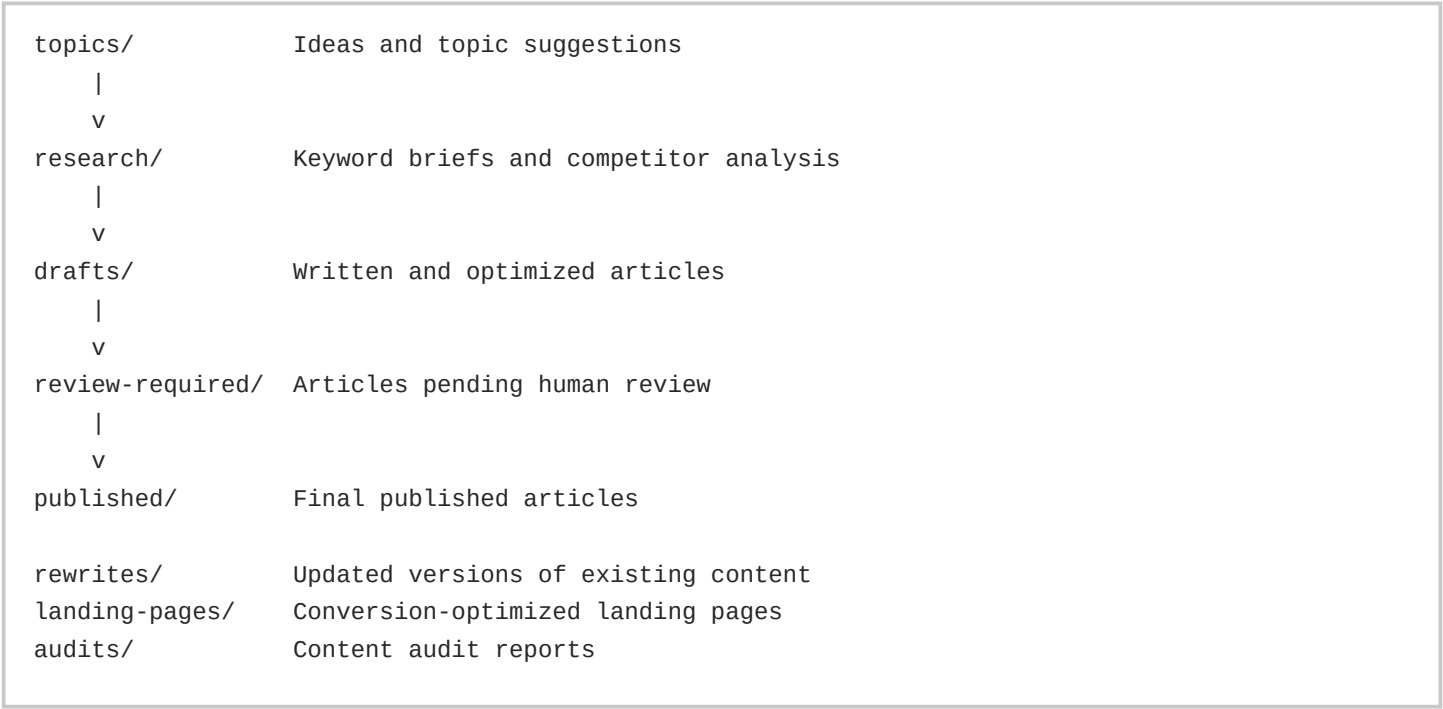
Edit these files in the context/ folder:

- **brand-voice.md** - Your tone and messaging style
- **style-guide.md** - Grammar and formatting rules
- **internal-links-map.md** - Your key pages for internal linking
- **features.md** - Your product/service features
- **seo-guidelines.md** - Your SEO requirements

Step 5: Install WordPress Plugin

Upload wordpress/seo-machine-woast-rest.php to your WordPress site's wp-content/mu-plugins/ directory. You're ready to go!

11. Content Pipeline Overview



12. Security and Safety

Full security audit results:

- All code is open-source and fully auditable
- No data is sent anywhere except your configured APIs (GA4, GSC, DataForSEO, WordPress)
- Credentials are stored locally in .env files (never committed to git)
- No telemetry, tracking, or hidden data collection
- WordPress PHP plugin uses proper input sanitization and permission checks
- Zero instances of dangerous functions (eval, exec, subprocess) in the codebase
- No hardcoded URLs, IPs, or backdoors
- No obfuscated code or hidden instructions
- All 21 Python dependencies are well-known, reputable packages
- The /scrub command actively defends against hidden Unicode manipulation

VERDICT: Safe to run. No security risks found.

13. Key Takeaways

1. SEO Machine is a free, open-source tool with no hidden costs or catches.
2. It provides an end-to-end pipeline from research to WordPress publishing.
3. 11 AI agents automatically optimize every article for SEO.
4. Integrates with real analytics data (GA4, GSC) for data-driven decisions.
5. One operator can produce significantly more content than manual workflows.
6. For agencies publishing 100 blogs/day, it dramatically reduces cost and time.
7. Customize once (brand voice, style guide) and all content follows your standards.
8. Safe, secure, and built by a credible SaaS founder (Craig Hewitt of Castos).
9. ~2,900 GitHub stars and coverage by Search Engine Land and other publications.
10. No manipulative code, backdoors, or hidden data collection of any kind.